



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

SENIOR CERTIFICATE EXAMINATIONS/ NATIONAL SENIOR CERTIFICATE EXAMINATIONS

AGRICULTURAL SCIENCES P2

MAY/JUNE 2026

MARKING GUIDELINES

MARKS: 150

These marking guidelines consist of 12 pages.

SECTION A**QUESTION 1**

1.1	1.1.1	D ✓✓	(10 x 2)	(20)
	1.1.2	D ✓✓		
	1.1.3	B ✓✓		
	1.1.4	B ✓✓		
	1.1.5	A ✓✓		
	1.1.6	C ✓✓		
	1.1.7	B ✓✓		
	1.1.8	A ✓✓		
	1.1.9	D ✓✓		
	1.1.10	C ✓✓		
1.2	1.2.1	C ✓✓	(5 x 2)	(10)
	1.2.2	H ✓✓		
	1.2.3	A ✓✓		
	1.2.4	I ✓✓		
	1.2.5	G ✓✓		
1.3	1.3.1	Intermediaries/middlemen/agents/brokers ✓✓	(5 x 2)	(10)
	1.3.2	Skilled/semi-skilled labour ✓✓		
	1.3.3	Epistasis ✓✓		
	1.3.4	Heredity ✓✓		
	1.3.5	Estimated breeding value/EBV ✓✓		
1.4	1.4.1	Entrepreneurship ✓	(5 x 1)	(5)
	1.4.2	Inventory ✓		
	1.4.3	Qualitative ✓		
	1.4.4	Alleles ✓		
	1.4.5	Electroporation ✓		

TOTAL SECTION A: 45

SECTION B**QUESTION 2: AGRICULTURAL MARKETING AND MANAGEMENT****2.1 Marketing functions****2.1.1 Identification of the marketing functions in****PICTURE A** - Storage ✓

(1)

PICTURE D - Transport/transportation ✓

(1)

2.1.2 ONE guideline for packaging

- Provides information about the product/identification/labelling ✓
- Convenient for handling/containment ✓
- Biodegradable/recyclable ✓
- Free from health risks/chemicals/foreign objects ✓
- Protection against mechanical damage ✓

(Any 1) (1)

2.1.3 TWO advantages of processing

- Increases the value of the product ✓
- Higher income for farmers ✓
- Overcome over-supply of the product ✓
- Allows easier packaging and handling of product/easy to transport/convenience ✓
- Provides job opportunities ✓
- Ensures the availability of product throughout the year ✓
- Reduces spoilage/wastage/perishability ✓
- Longer shelf life ✓
- Ensures food security ✓
- Improved food safety ✓
- Makes the products more appealing to the consumer ✓

(Any 2) (2)

2.2 Elasticity of demand**2.2.1 Indication of the form of elasticity****(a) PICTURE B** - Inelastic ✓

(1)

(b) PICTURE A - Elastic ✓

(1)

2.2.2 Reason for the elasticity

The change in price resulted in a huge change in demand ✓✓

(2)

2.2.3 ONE factor that led to inelasticity

- Nature/usefulness of the product/habitual consumption □
- Lack of alternatives/availability of products □
- Time period/season ✓
- Consumers income ✓

(Any 1) (1)

2.3 Price determination**2.3.1 Identification of an equilibrium price**

R80 ✓

(1)

2.3.2 Explanation of the relationship between price and the supply

- The higher the price the higher the supply of mangoes ✓✓
- The lower the price the lower the supply of mangoes ✓✓
- The price and supply are proportionally related to one another ✓✓

(Any 1)

(2)

2.3.3 TWO factors affecting supply of mangoes

- Price ✓
- Competitive products/number of producers ✓
- Environmental conditions/pests and diseases ✓
- Political instability ✓
- Expectations of future price change ✓
- Demand for the product ✓
- Technology ✓
- Research/knowledge of production ✓
- Production costs ✓
- Subsidies/taxation ✓
- Possibilities of increasing the supply of products ✓
- Seasonality of the product ✓
- Legislation ✓
- Infrastructure ✓

(Any 2)

(2)

2.3.4 TWO factors that hamper the marketing of mangoes

- Perishability/spoilage ✓
- Low value in relation to high volume ✓
- Lack of control over production/diversity in production ✓
- Wide distribution of products/transport costs ✓
- Standardisation ✓
- Seasonal changes ✓
- Poor infrastructure/accidents/theft ✓
- Lack of capital ✓
- Competition ✓

(Any 2)

(2)

2.4 Description for fresh produce markets

Fruits and vegetables are marketed in the larger centres/municipal markets ✓
by agents for commission ✓

(2)

2.5 Marketing chain**2.5.1 Identification of**

- (a) **The processing plant** - Abattoir/butchery ✓ (1)
- (b) **Retailer** - Butchery ✓ (1)
- (c) **Producer**- Farm ✓ (1)

2.5.2 TWO ways to streamline and improve the agri-business chain

- Improving infrastructure ✓
- Cooling/storage facilities ✓
- Improving communication ✓
- Use of the latest technology ✓
- Make use of internet/social platforms ✓
- Combining loads/partnerships ✓
- Process products near the farm ✓
- Increased governmental control over imports ✓ (Any 2) (2)

2.6 Entrepreneurship

- D ✓ (1)
- C ✓ (1)
- A ✓ (1)
- B ✓ (1)

2.7 Business plan**2.7.1 ONE aspect that should be included in**

- (a) **The title page**
 - Name of the business/person ✓
 - Logo ✓
 - Address ✓
 - Contact details of the business/person ✓ (Any 1) (1)
- (b) **Human resource plan**
 - Number and type of employees ✓
 - Competencies and skills needed ✓
 - Recruitment process/capacity building of staff ✓ (Any 1) (1)

2.7.2 ONE item that should be included in a good business plan

- Name of business/partners/partnership ✓
- Business strategy ✓
- Production/operational plan ✓
- Management plan/time management plan ✓
- Sales and marketing plan ✓
- Financial plan ✓
- Reasons for success of business ✓
- SWOT analysis ✓ (Any 1) (1)

2.8 SWOT Analysis**Indication of the component of SWOT analysis**

A	Threat ✓	(1)
B	Weakness ✓	(1)
C	Strength ✓	(1)
D	Opportunity ✓	(1)
		[35]

QUESTION 3: PRODUCTION FACTORS**3.1 The law of diminishing returns****3.1.1 Identification of the range where each of the following occurs:**

- | | | |
|-----|---------------------------------------|-----|
| (a) | Diminishing returns - B to C ✓ | (1) |
| (b) | Negative returns - C to D ✓ | (1) |
| (c) | Increasing returns - A to B ✓ | (1) |

3.1.2 The letter representing a point of maximum production

C ✓	(1)
-----	-----

3.1.3 ONE method to improve land productivity

- | | | |
|--|---------|-----|
| <ul style="list-style-type: none"> Improving soil health by adding fertiliser/adding organic matter ✓ Optimizing water management ✓ Selecting appropriate/improved cultivars/crops ✓ Implementing efficient scientific farming practices/reduce tillage/crop rotation/intercropping ✓ Use of improved technology/precision farming ✓ Land use optimization/putting together uneconomic units ✓ Infrastructure development ✓ Research/extension/education ✓ | (Any 1) | (1) |
|--|---------|-----|

3.2 Economic characteristic of land

3.2.1	Land is limited/scarcity of land/availability ✓	(1)
-------	---	-----

- | | | | |
|-------|---|---------|-----|
| 3.2.2 | <ul style="list-style-type: none"> Land appreciates in value ✓ Land can be bought or sold ✓ | (Any 1) | (1) |
|-------|---|---------|-----|

3.3 Labour problems**3.3.1 ONE cause that may result in the loss of farm labourers**

- | | | |
|---|---------|-----|
| <ul style="list-style-type: none"> Unfavourable working conditions/working hours ✓ Lower wages/salaries ✓ Poor living conditions ✓ Lack of training ✓ Better opportunities in the industries ✓ | (Any 1) | (1) |
|---|---------|-----|

3.3.2 Explanation of how ill-health threatens the stability of the labour force and production on the farm

The ill-health of the labourers causes absenteeism/loss of skills and experience ✓ and subsequently drop in agricultural production/income ✓

(2)

3.3.3 TWO implications of the Labour Relations Act

- Involvement in decision making ✓
- Right to join a union/strike ✓
- Economic development ✓
- Prevention of unfair labour practice ✓
- Provision of procedures for the resolution of labour disputes ✓
- Ensuring social justice and fair working conditions ✓ (Any 2)

(2)

3.4 Labour contract**3.4.1 Item in the contract serving as a proof of an agreement**

Signatures/signing/writing names ✓

(1)

3.4.2 ONE other item to be included in a contract

- Remuneration details/wages ✓
- Place of work ✓
- Job description ✓
- Termination of employment ✓
- Hours of work ✓
- Leave benefits ✓
- Period/duration of employment ✓
- Fringe benefits ✓
- Medical benefits ✓
- Mode of payment ✓
- Details of the employer and employee ✓

(Any 1)

(1)

3.5 Capital**3.5.1 Indication of the types of capital**

A - Production/floating/working capital ✓

(1)

B - Movable capital ✓

(1)

3.5.2 Source of credit used to purchase fixed capital

Land bank/Commercial banks ✓

(1)

3.5.3 Deduction of the problem of movable capital

- Risk factor ✓
- Depreciation ✓
- Expensive/mechanical/maintenance ✓

(Any 1)

(1)

3.6 Budget

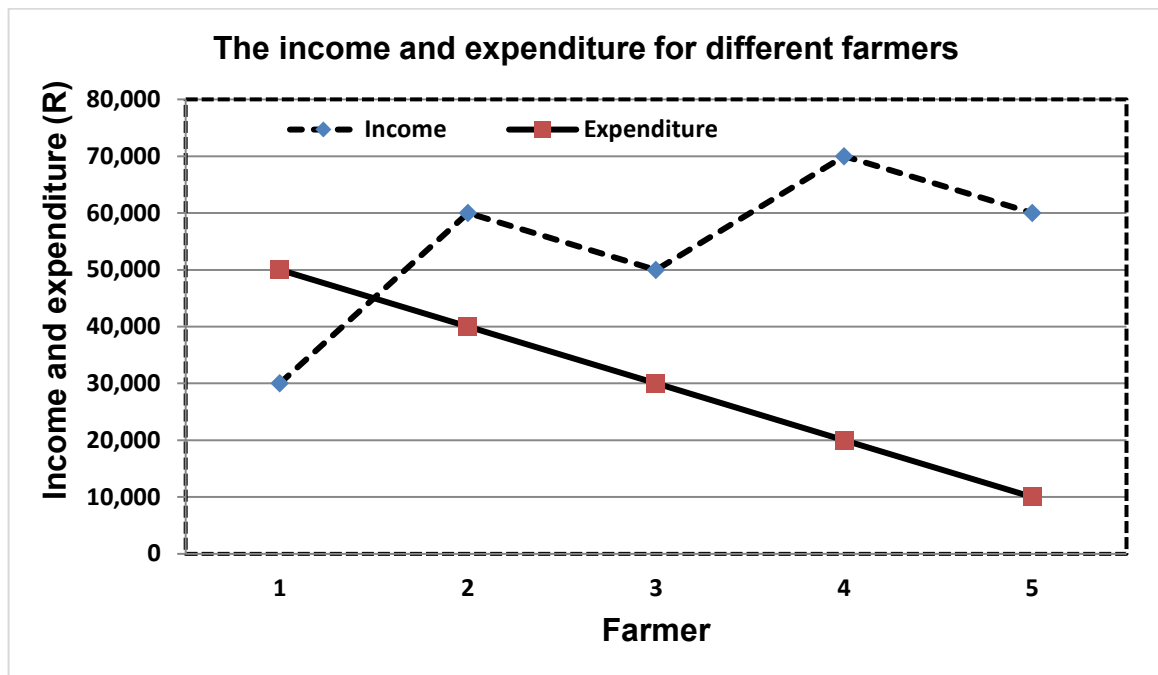
- 3.6.1 **Name of the financial record**
Whole farm budget/budget ✓ (1)
- 3.6.2 **Identification of an example of**
(a) **Fixed costs** - Insurance ✓ (1)
(b) **Variable costs** - Feeds/fuel/pesticides/electricity/seeds/wages ✓ (1)
- 3.6.3 **Calculation of the profit/loss for the farming business**
Profit/loss = Total income – Total expenditure ✓
= R280 000 – R133 400 ✓
= R146 600 ✓ (3)

3.7 Management

- 3.7.1 **Type of risk encountered by the manager**
Market/price/financial risk ✓ (1)
- 3.7.2 **One risk management strategy**
• Risk sharing ✓
• Diversification ✓ (Any 1) (1)
- 3.7.3 **ONE principle of management**
• Planning/setting goals ✓
• Implementation/coordinating ✓
• Control/monitoring ✓
• Decision making ✓
• Organisation ✓
• Communication/motivation/leading/directing ✓ (Any 1) (1)
- 3.7.4 **Explanation of how competition affects farming business**
With more competition in the market, more products will be produced ✓
resulting in better quality, a wider variety of products resulting in price
fluctuation in the market ✓ (2)

3.8 Line graph

3.8.1 Line graph

**CRITERIA/RUBRIC/MARKING GUIDELINES**

- Correct heading (both variables included) ✓
- X-axis: Correctly calibrated with label (Farmer) ✓
- Y axis: Correctly calibrated with label (Income and expenditure) ✓
- Correct units (R) ✓
- Line graph ✓
- Accuracy (80%+ correctly plotted) ✓

(6)
[35]**QUESTION 4: BASIC AGRICULTURAL GENETICS**

4.1 Chromosomes

4.1.1 Identification of the examples

- (a) Heterozygous alleles - Gg ✓ (1)
- (b) Homozygous recessive alleles - rr/bb ✓ (1)

4.1.2 Naming the fixed position

Locus ✓ (1)

4.2 Patterns of inheritance**4.2.1 Identification of the patterns of inheritance**

- (a) **PICTURE A** - Incomplete dominance ✓ (1)
- (b) **PICTURE B** - Co-dominance ✓ (1)

4.2.2 Justification for co-dominance

- Colour for both parents is represented in the offspring ✓
- Both red and white colours/roan are present in the offspring ✓
- The offspring has the phenotype of both parents ✓
- Both red and white are equally dominant ✓
- No intermediate or mixture of colour is formed ✓ (Any 1) (1)

4.3 Cross between a black bull and white cow**4.3.1 Identification of the type of crossing**

Monohybrid crossing ✓ (1)

4.3.2 Punnett square

Gametes	B	B
b	Bb	Bb
b	Bb	Bb

Marking rubric

- Correct gametes of male parent ✓
- Correct gametes of female parent ✓
- Correct genotype of the offspring ✓
- Correct punnet square ✓ (4)

4.3.3 Calculation of the percentage black offspring in F₁ generation

$$\frac{4}{4} \times 100 \checkmark$$

$$= 100\% \checkmark \quad (2)$$

4.4 Patterns of inheritance**4.4.1 Indication of the type of inheritance**

Polygenic inheritance ✓ (1)

4.4.2 Calculation of the maximum milk yield for the dairy cow per day

- 40+(5x6) litres ✓
- 40+30 litres ✓
- = 70 litres ✓ (3)

4.4.3 Genotype that can produce 60 litres of milk per day

- PPQQrr ✓
- ppQQRR ✓
- PpQqRR ✓
- PPqqRR ✓
- PPQqRr ✓
- PpQQRr ✓

(Any 1) (1)

4.5 Variation**4.5.1 The name of the phenomenon**

Variation ✓

(1)

4.5.2 ONE genetic cause of the variation

- Mutation ✓
- Meiosis/crossing over/recombination of genes ✓
- Random fertilisation ✓

(Any 1) (1)

4.5.3 Explanation of the statement

- Variation in traits within a population allows breeders to choose individuals ✓ that are more productive, disease-resistant, or have other desirable characteristics ✓
- Variation provides the raw material for breeders to identify and select individuals ✓ with desirable traits for improving crops and livestock ✓

(Any 1) (2)

4.6 Breeding and selection**4.6.1 Indication of a breeding system**

Upgrading ✓

(1)

4.6.2 The number of generations to develop a breed with 87% characteristics of a Hereford - 3 (three) ✓

(1)

4.6.3 Deduction of ONE disadvantages of upgrading

- Improvement is slow ✓
- The offspring can never breed 100% pure ✓
- Expensive because bulls are renewed ✓
- Bull calves of the first few generations cannot be sold as breeding stock ✓

(Any 1) (1)

4.6.4 ONE related breeding system

- Inbreeding ✓
- Line breeding ✓

(Any 1) (1)

4.7 Heritability

- 4.7.1 **Identification of a characteristic the farmer will mostly not consider**
Fleece weight ✓ (1)
- 4.7.2 **Explanation of the effectiveness**
Meat tenderness has a heritability of 80% ✓ and therefore, there is a high possibility that the characteristic can be passed to the offspring for the improvement of the herd ✓ (2)
- 4.7.3 **Type of selection**
Artificial selection ✓ (1)

4.8 Plant improvement

- 4.8.1 **Identification of process of plant improvement**
Genetic modification (GM) /engineering ✓ (1)
- 4.8.2 **Name of the technique used**
Bacterial carriers/Use of *Agrobacterium tumefaciens* ✓ (1)
- 4.8.3 **The purpose of using genetic modification**
Used to develop insect resistant fruit/to increase insect resistance in plants ✓ (1)
- 4.8.4 **ONE other use of genetic modification**
- Improve disease/pest resistance ☐
 - Improve herbicide tolerance ☐
 - Improve resistance to harsh environmental conditions/salinity ☐
 - Improve nutritional value of the crop ☐
 - Improve yield ☐
 - Longer shelf life ☐ (Any 1) (1)
- 4.8.5 **ONE potential health risks of GMO**
- It can lead to the development of food allergies ✓
 - It may cause long term negative health effects ✓ (Any 1) (1)
- [35]**

TOTAL SECTION B: 105
GRAND TOTAL: 150